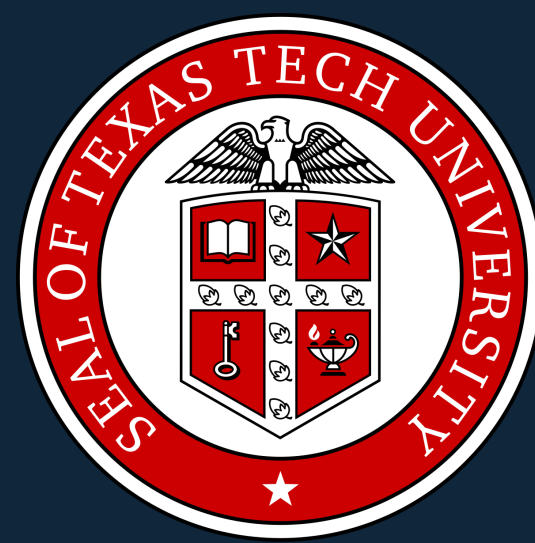




Second Order Reduction for the Helfrich Energy

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Motivation

Primarily motivated by the **Willmore energy**, recent decades have seen a surge in interest in fourth order geometric variational problems for surfaces. In 1973 W. Helfrich, based on previous studies on liquid crystallography, proposed an extension of the Willmore energy to understand the morphology of **bilipid membranes**. Contrary to the Willmore case, the Helfrich energy is no longer conformally invariant—an essential property in the study of the variational problems associated to the Willmore energy.

§1 The Helfrich Energy

For an **embedding** $X : \Sigma \longrightarrow \mathbb{R}^3$ the **Helfrich energy** is

$$\mathcal{H}[\Sigma] := \int_{\Sigma} (H + c_o)^2 d\Sigma,$$

where H is the **mean curvature** and $c_o \in \mathbb{R}$ is a constant known as the spontaneous curvature. This constant accounts for the asymmetry between the two layers of the membrane.

The **Euler-Lagrange equation** associated to $\mathcal{H}$ is the fourth order nonlinear elliptic PDE

$$\Delta H + 2(H + c_o)(H(H - c_o) - K) = 0. \quad (1)$$

Surfaces with constant mean curvature (CMC) $H \equiv -c_o$ are trivial solutions.

Theorem ([2] Palmer & P., 2022)

An axially symmetric topological disc critical for $\mathcal{H}$ is either a CMC $H \equiv -c_o$ surface or, after normalizations,

$$H + c_o = -\frac{\nu_3}{z}, \quad (2)$$

holds on Σ . (Here $\nu_3 := \nu \cdot E_3$ is the vertical component of the unit normal to Σ .)

§2 The Reduced Membrane Equation

Equation (2) is the Euler-Lagrange equation for the action

$$\mathcal{G}[\Sigma] := \int_{\Sigma} \frac{1}{z^2} d\Sigma - 2c_o \int_{\Omega} \frac{1}{z^2} dV,$$

where $\Omega \subset \mathbb{R}^3$ is the algebraic volume enclosed by the surface Σ .

Wherever $z \neq 0$ holds, the surface Σ can be regarded, after a conformal change of metric, as a surface in the **hyperbolic space** $\mathbb{H}^3$ and the functional $\mathcal{G}$ can be expressed as

$$\mathcal{G}[\Sigma] = \tilde{\mathcal{A}}[\Sigma] - 2c_o \tilde{\mathcal{U}}[\Sigma] := \int_{\Sigma} d\tilde{\Sigma} - 2c_o \int_{\Omega} |z| d\tilde{V},$$

where $\tilde{\mathcal{A}}$ is the **hyperbolic area** and $\tilde{\mathcal{U}}$ is the **hyperbolic gravitational potential energy**.

Regardless of the topological type and whether the surface is axially symmetric or not, we have:

Proposition ([2] Palmer & P., 2022)

A sufficiently regular surface satisfying the reduced membrane equation (2) must also satisfy (1).

§3 Bifurcation Analysis

Among axially symmetric disc type solutions of the reduced membrane equation (2) there exists one having a **horizontal tangent plane** along its boundary circle, $X(\Sigma_o)$. (See the surface in the center of Figures 1-2.)

Theorem ([5] Palmer & P., 2024)

The surface $X(\Sigma_o)$ is embedded in a one parameter family of axially symmetric solutions of (2), parameterized by c_o , all of which share the same boundary circle.

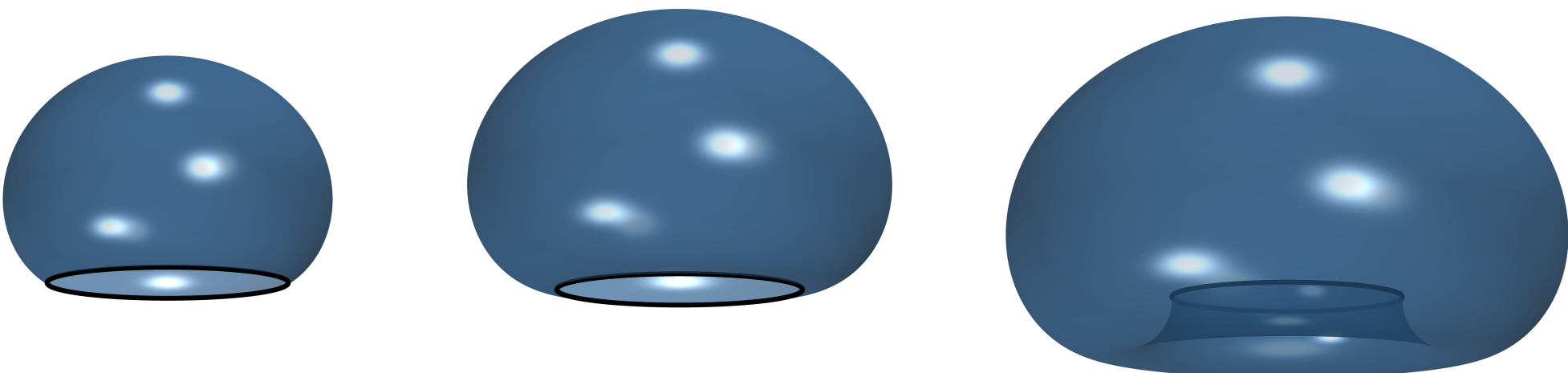


Figure 1. Axially symmetric family of surfaces satisfying (2) with the same boundary circle.

Theorem ([4] Palmer & P., 2024)

The surface $X(\Sigma_o)$ and its subdomains are stable for the functional $\mathcal{G}$, while superdomains are unstable.

This transition between stability and instability suggests that a bifurcation may occur. Applying Crandall and Rabinowitz's theory of 1971 for **simple eigenvalues** the existence of a **pitchfork bifurcation** is shown. The procedure is similar to that used by Koiso, Palmer, and Piccione in 2017 for surfaces with CMC.

Theorem ([5] Palmer & P., 2024)

At the surface $X(\Sigma_o)$ a non-axially symmetric branch of solutions of (2) bifurcates while maintaining the same circular boundary.

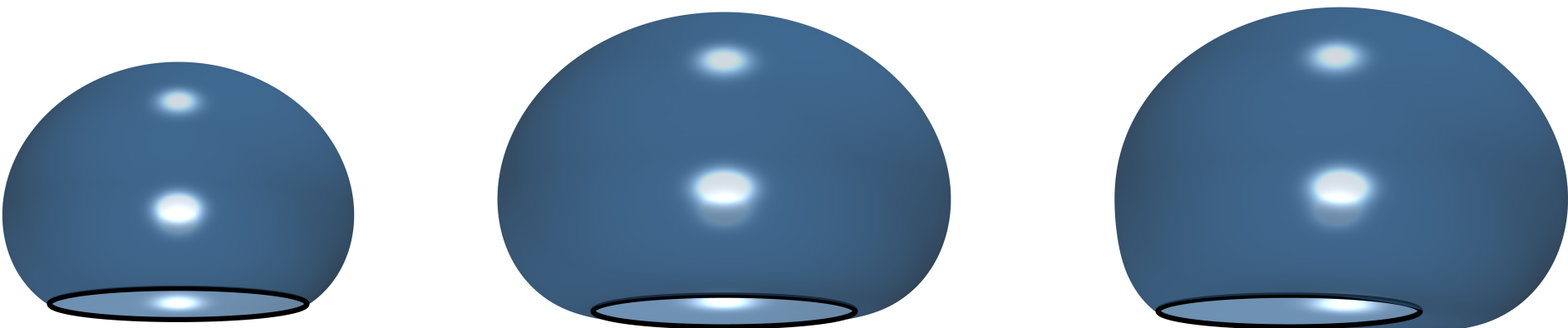


Figure 2. Bifurcating branch of surfaces satisfying (2) with the same boundary circle.

§4 The Regularized Functional $\mathcal{G}_R$

Assume that the surface Σ is properly embedded in $\mathbb{R}_+^3 = \mathbb{R}^3 \cap \{z \geq 0\}$ and that its boundary $\partial\Sigma$ lies in the plane $\{z = 0\}$, meeting this plane orthogonally. Such a surface can be understood as a surface in $\mathbb{H}^3$ approaching the **ideal boundary** $\partial_{\infty}\mathbb{H}^3$ at a **right angle**.

Both the hyperbolic area $\tilde{\mathcal{A}}$ and the hyperbolic gravitational potential energy $\tilde{\mathcal{U}}$ of Σ are infinite, but they possess certain regularizations:

$$\mathcal{A}_R[\Sigma] := \lim_{\epsilon \rightarrow 0^+} \left(\int_{\Sigma_{\epsilon}} \frac{1}{z^2} d\Sigma + \frac{1}{\epsilon} \oint_{\partial\Sigma_{\epsilon}} \star dz \right),$$

and

$$\mathcal{U}_R[\Sigma] := \lim_{\epsilon \rightarrow 0^+} \left(\int_{\Omega_{\epsilon}} \frac{1}{z^2} dV - \frac{1}{\epsilon} |\Omega \cap \{z = \epsilon\}| \right),$$

where $\Sigma_{\epsilon} := \Sigma \cap \{z \geq \epsilon\}$, $\Omega_{\epsilon} = \Omega \cap \{z \geq \epsilon\}$, and $|\Omega \cap \{z = \epsilon\}|$ represents the Euclidean area of $\Omega \cap \{z = \epsilon\}$.

We then consider the **regularized functional**

$$\mathcal{G}_R[\Sigma] := \mathcal{A}_R[\Sigma] - 2c_o \mathcal{U}_R[\Sigma],$$

and show its relation with the Helfrich energy $\mathcal{H}[\Sigma]$:

Theorem ([3] Palmer & P., 2025)

The regularized functional $\mathcal{G}_R$ of Σ satisfies

$$\mathcal{G}_R[\Sigma] = \int_{\Sigma} \left(\tilde{H} + c_o z \right)^2 d\tilde{\Sigma} - \mathcal{H}[\Sigma],$$

where $\tilde{H}$ is the hyperbolic mean curvature of Σ .

The specific case $c_o = 0$ answers a question raised in a 2010 paper by Alexakis and Mazzeo, regarding if a simple connection exists between the **renormalized area** $\mathcal{A}_R$ and the **Willmore energy** for **non-minimal** surfaces in $\mathbb{H}^3$.

Consider variations tangent to $\{z = 0\}$ and preserving the contact angle with this plane.

Proposition ([3] Palmer & P., 2025)

A surface is critical for the regularized functional $\mathcal{G}_R$ if and only if the reduced membrane equation (2) holds and $\partial_n H = 0$ along the boundary. (Here n is the outward pointing conormal to $\partial\Sigma$.)

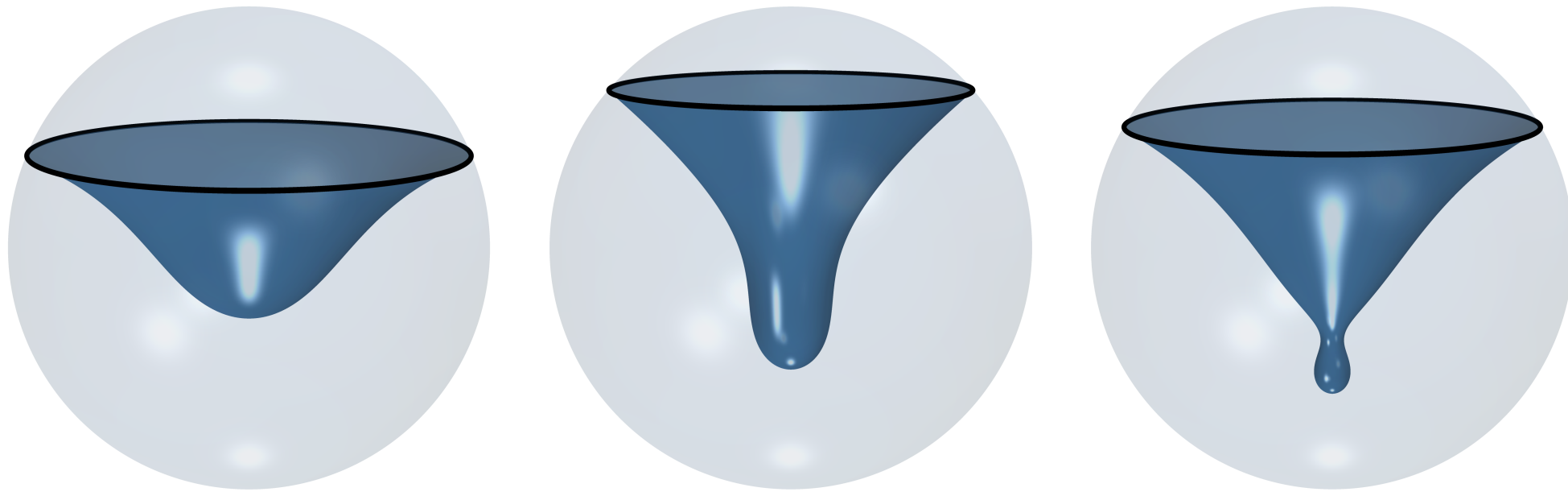


Figure 3. Equilibria for the regularized functional $\mathcal{G}_R$.

§5 Closed Helfrich Surfaces

Let Σ be a **closed genus zero** surface and $X : \Sigma \longrightarrow \mathbb{R}^3$ an **axially symmetric** embedding. If the surface has CMC, due to classical results of Hopf and Alexandrov, it must be a round sphere. Assume that the surface has **non-constant mean curvature**. We then have:

Theorem ([1] López, Palmer & P., 2026)

The surface is critical for the Helfrich energy $\mathcal{H}$ if and only if, up to normalizations, the reduced membrane equation (2) holds and the rescaling condition

$$\int_{\Sigma} (H + c_o) d\Sigma = 0,$$

is satisfied.

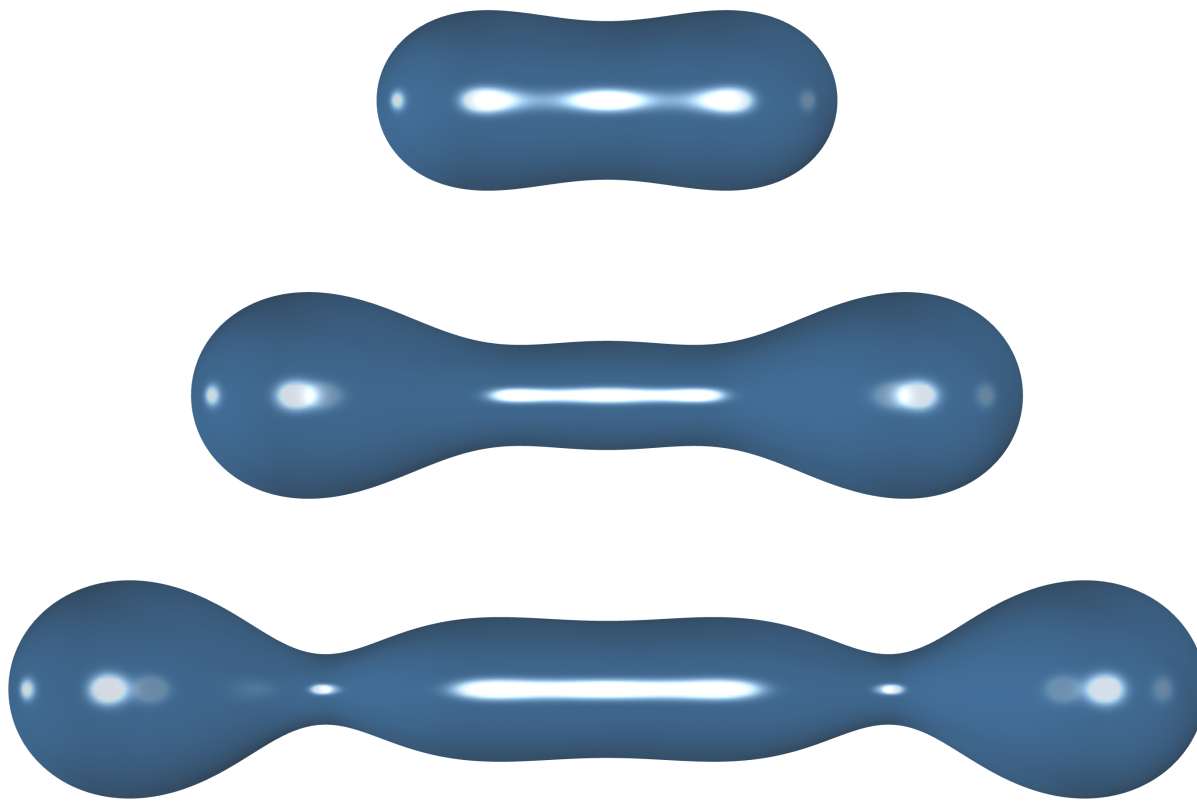


Figure 4. Axially symmetric closed Helfrich surfaces of genus zero with non-constant mean curvature.

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